

## **Mini Case Study: Real-life Fraud Detection Scenario**

### **1 Case Background**

The dataset used in this report is derived from real transaction data provided by Vesta Corporation. Exploratory Data Analysis “EDA” methodology is used to conduct preliminary analysis of the data and use machine learning models to identify potentially fraudulent transactions, which will help organizations reduce fraud losses and improve the reliability of their payment systems, thereby providing consumers with a safer and more trustworthy payment experience.

### **2 Summary Description of Dataset**

This dataset predicts whether an online transaction is fraudulent, with the target variable “isFraud” indicating fraud (1) or non-fraud (0). It contains 101 variables across four data types: 19 character, 59 integer, 14 logical, and 9 numeric variables. The dataset includes 100,000 observations, with 11,318 (11.3%) labeled as fraud and 88,682 (88.7%) as non-fraud (Appendix Figure 1).

The dataset is divided into two parts: Transaction Features and Identity Features. The Transaction Features include details such as transaction amount (“TransactionAMT”), product code (“ProductCD”), payment card info (“card1” – “card6”), address (“addr1”, “addr2”), email domains (“P\_emaildomain”, “R\_emaildomain”), time difference (“TransactionDT”), and several derived features (C1-C14, D1-D15, M1-M9, Vxxx) related to counts, time differences, and matches. The Identity Features provide information about the device and network, including “DeviceType”, “DeviceInfo”, and masked features (“id\_12” – “id\_38”), useful for analyzing user behavior and connectivity.

### **3 Exploratory Data Analysis (EDA)**

#### **3.1 Univariate Analysis**

In order to show the potential characteristics of the data and its patterns, I conducted Univariate Analysis. Based on the results of data description, I chose several important features to analyze, including “TransactionAMT”, “TransactionDT” and “card1”-“card6”. By observing their histograms and box plots, I came up with the following conclusions: the data distribution of “card1” is more uniform, indicating that users use more diverse payment card information; “TransactionDT” shows a left-skewed trend, indicating that the distribution of transaction time is more centralized (); the distribution of “TransactionAMT” is also left-skewed, and most of the transaction amounts are small, but there are a few extremely large transaction The distribution of “TransactionAMT” is also left-skewed, most of the transaction amounts are small, but there are a few extremely large transactions, which may be a potential signal of fraudulent transactions (Figure 1).

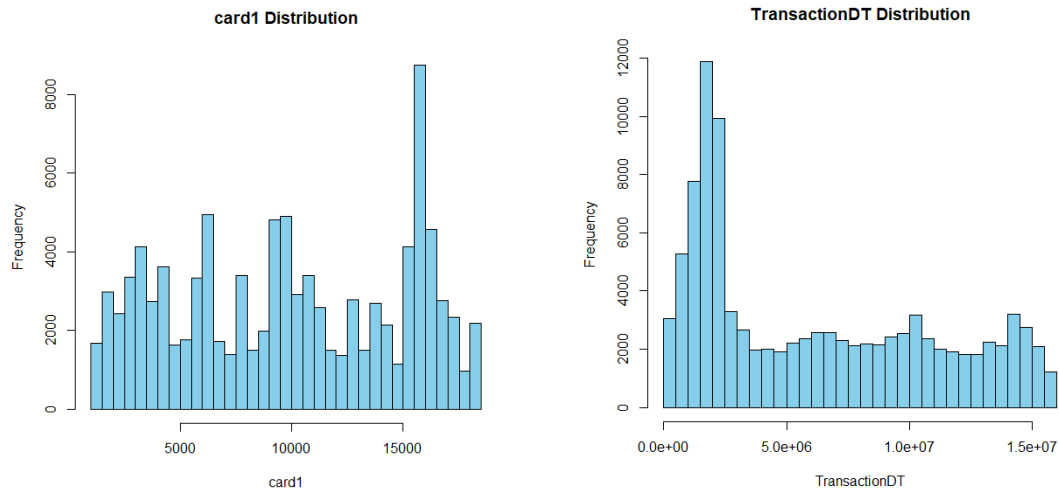


Figure 1 card1 Distribution and TransactionDT Distribution

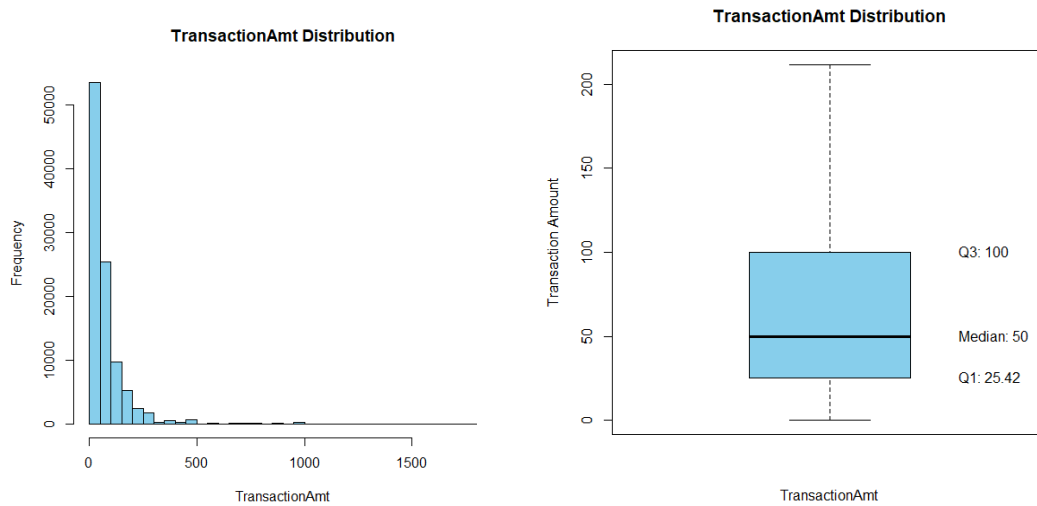


Figure 2 Histogram and Boxplot of TransactionAMT

In the histogram of isfraud (Appendix Figure 2), the ratio of positive and negative classes is 1:8, which is a data imbalance problem. In order to ensure that the model can accurately identify fraudulent transactions, the data imbalance problem needs to be dealt with subsequently. In addition, during the modeling process, it is necessary to ensure that the predictive effect of the model is not affected.

### 3.2 Bi-/Multi-variate Analysis

To begin the analysis, I computed the correlation coefficients between all pairs of numerical variables, generating a correlation matrix that quantifies the linear relationships among them (Appendix Figure 3). The matrix reveals that several variable groups (C1-C2, C4-C13, D1-D14, V310-V314) exhibit strong correlations, with absolute coefficients greater than 0.8. These high correlations can inflate standard errors and increase the risk of overfitting in the model. To mitigate these issues, it is essential to apply dimensionality reduction to the highly correlated variable sets, thereby simplifying the model and enhancing its predictive performance.

## 4 Data Cleaning

#### 4.1 Missing Data

I first removed variables with more than 70% missing values to streamline the analytical process. These variables, due to their high proportion of missing data, were deemed uninformative and could negatively impact model performance if left in the dataset. Removing such variables simplified the dataset, allowing for more reliable analysis and reducing complexity.

#### 4.2 Feature Selection

During the feature selection process, I identified multiple variables with an absolute correlation coefficient greater than 0.8. To address this high multicollinearity, I retained C1 and C3 to represent the C variables, D9 and D15 to represent the D variables, and V311 and V314 to represent the V variables. This process effectively reduces multicollinearity, enhancing the stability and performance of the model. By reducing the number of highly correlated variables, the model is less likely to overfit and is more likely to generalize well to new data. After this adjustment, the updated correlation matrix (Appendix Figure 4) shows that the majority of the remaining variables are not highly correlated, ensuring a more robust dataset for modeling.

#### 4.3 Deletion

For deletion, categorical variables with too many rare categories were categorized or combined to prevent overfitting and computational instability. Numerical variables with little impact on the model were removed to improve model performance. All categorical variables were converted into integer labels, while binary logistic variables were represented as 0s and 1s. To identify features significantly associated with the target variable "isFraud," Fisher's test and chi-square test were used.

#### 4.4 Imputation

##### 4.4.1 Mean Imputation

For continuous variables, missing values were replaced with the mean value of the respective variable..

##### 4.4.2 Mode Imputation:

For categorical variables, missing values and "NA" were imputed with the mode of the variable. The mode was calculated by excluding missing values and "NA" to ensure an accurate representation of the most frequent value in the dataset. This method helps preserve the integrity of categorical data while minimizing the impact of missing values.

#### 4.5 Capping/Flooring outlier

For data points exceeding the third quartile or below the first quartile, values are capped or floored accordingly. However, for key variables like transaction amount, which play a critical role in fraud detection and contain many large values, capping at the third quartile could overlook important cases. Therefore, these variables are handled separately to preserve essential information. Additionally, variables with very few valid data points are retained in their original form to avoid losing valuable data during outlier treatment.

### 5 Feature Engineering

#### 5.1 Sequence Processing

For the "TransactionID" feature, large, random numerical values were replaced with a simple sequence to reduce complexity. This transforms the feature into an ordinal identifier, helping the model treat it as a categorical variable rather than a continuous one, thereby minimizing noise and improving clarity.

## 5.2 Handling Special Outliers

Based on the practical significance of the variable, TransactionAmt is identified as a key feature for detecting fraud. Capping it could result in the loss of important data and negatively impact model performance. Therefore, this report retains the original data of TransactionAmt and create a new variable, RedFlag, which marks values above the 75th percentile as 1. This approach ensures that valuable information is preserved for subsequent model predictions.

## 5.3 Normalization

To address discrepancies in units among numerical variables, I standardized all numerical features using the z-score method to prevent unequal weighting during model prediction. Specifically, the time feature, which varies greatly in scale, is normalized using min-max scaling to a range between 0 and 1. This ensures that the time feature contributes proportionally to the model without disproportionately affecting other features due to its large numerical range.

## 5.4 Balanced datasets

As a solution to the imbalance in data, I implemented the SMOTE (Synthetic Minority Over-sampling Technique) algorithm for conducting oversampling. SMOTE creates artificial samples through a process of interpolation between minority class samples and adjacent samples. By setting K to 5 and contemplating five closest neighbors to produce fresh samples, and Dup\_size to 6, we created six additional samples per sample of minority class. Following the utilization of SMOTE, the dataset comprised 88,682 samples (52.82%) for the "fraud" category and 79,226 samples (47.18%) for the "non-fraud" category, culminating in an even distribution of 160,000 samples, maintaining a class ratio of 1:1. (Appendix Figure 5).

# 6 Model Building

I focuses on two financial fraud detection models: random forests and neural networks. Random forests can effectively deal with high-dimensional data and class imbalance problems, and provide better model interpretability and support for feature analysis. Neural networks, on the other hand, have been chosen to explore their potential in fraud detection due to their strong learning ability in capturing complex nonlinear relationships.

## 6.1 The process of tuning parameters

During model training, I optimized the hyperparameters for both models. For the random forest model, I adjusted the mtry and ntree parameters. Through cross-validation, I chose mtry=15 and ntree=500. For the neural network model, I adjusted maxit (the maximum number of iterations), size (the number of units in the hidden layer), and decay (the rate of weight decay), which were set to maxit=600, size=20, and decay=0.01, respectively, in order to increase the convergence speed of the model and to reduce the over risk of overfitting. I compared the performance of the model before

and after, and found that the model has significantly improved in terms of accuracy, recall and F1 value.

## 6.2 Model Evaluation

In the model evaluation section, I use a variety of performance metrics to evaluate the performance of the models on the test set, including Accuracy, Recall, F1 Score, and ROC curves.

### (1) Random Forest

After the data imbalance problem is effectively dealt with, the model achieves an accuracy of 0.9678, showing good overall prediction ability. In terms of recall, the model achieves 0.9797, which is especially good in identifying positive class samples (fraudulent transactions). The F1 value is 0.9700, which indicates that the model achieves a good balance between precision and recall. From the confusion matrix, the model performs well in predicting negative class samples and also maintains a low error rate (0.0426) in predicting positive class samples (Appendix Figure 6). Meanwhile, the AUC is 0.9923, which further validates the model's excellent performance in distinguishing between positive and negative class samples (Figure 3).

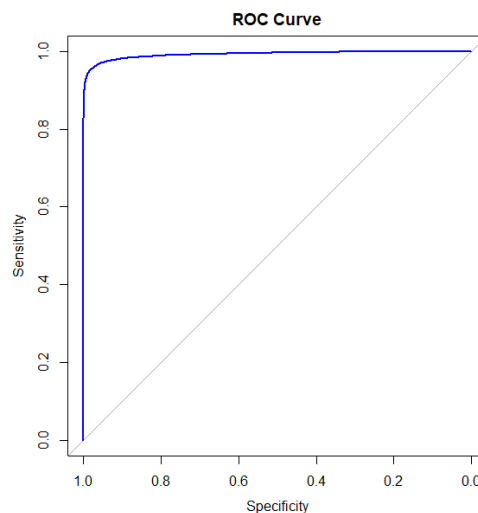


Figure 3 ROC of Random Forest

### (2) Neural Network

The training results show that the neural network model has an accuracy of 0.8779, indicating that it is able to correctly classify most of the samples (Appendix Figure 7). The recall is 0.8798, indicating that the model performs well in recognizing positive class samples (fraudulent transactions). The F1 value is 0.8858, reflecting the model's balance between precision and recall, which is able to effectively deal with the data imbalance problem. Meanwhile, the AUC is 0.9478 (Figure 4), indicating that the model is able to distinguish between positive and negative class samples better .

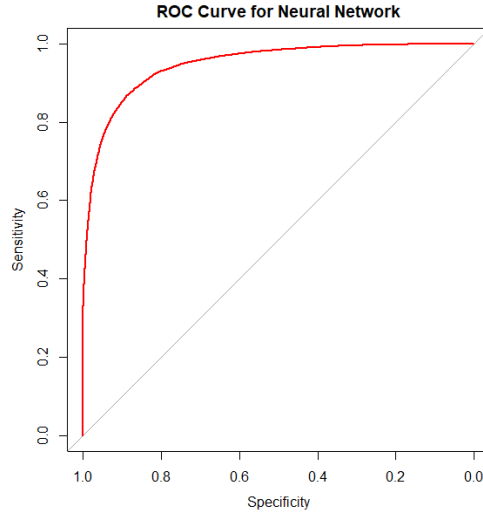


Figure 4 ROC of Neural Network

The training process of the neural network shows that the error rate gradually decreases as the number of training rounds increases, and the error rate decreases faster from round 1 to round 4, indicating that the model quickly learns the laws of the data in the initial optimization stage (Figure 5). Thereafter, the error rate tends to stabilize, and at the end of the 7th round of training, the error rate is close to 0.1, showing the stability and convergence of the model. In addition, the performance trends of the training and test sets are highly consistent, and the error rate of the test set decreases in tandem with that of the training set, indicating that the model does not undergo overfitting and is able to generalize to unseen data better .

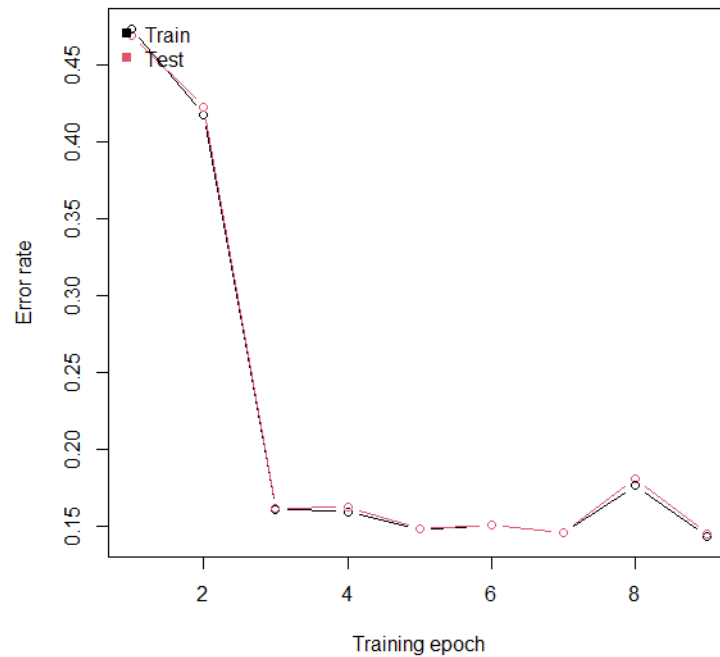


Figure 5 Training Epoch of Neural Network

### 6.3 Model Performance Comparison

According to the performance evaluation results of the random forest model and the neural network model, there are obvious differences in the performance of the two models on different indicators. Specifically, the random forest model performs well in key metrics such as accuracy (0.9678), recall (0.9797), F1 value (0.9699) and AUC (0.9923), especially in recognizing positive class samples (fraudulent transactions), which shows high recognition ability and low error rate. In comparison, the neural network model has an accuracy of 0.8779, a recall of 0.8798, an F1 value of 0.8858, and an AUC of 0.9478, which is inferior to the random forest model in overall performance. In addition, the neural network shows better convergence during the training process, and the error rate is gradually reduced to nearly 0.09, but it is relatively weak in the prediction ability of positive class samples. Overall, the random forest has a better performance in improving the recognition rate of fraudulent transactions.

## 7 Feature Importance and Redflag

### 7.1 Feature Importance

Based on the feature importance analysis of the random forest model, C1, id\_02 and card6 are identified as the most important features for fraud detection, showing the key role of these features in predicting fraudulent transactions (Figure 6). Specifically, the highest importance of feature C1 (223.88) indicates that it plays a decisive role in determining fraudulent transactions in the model. This feature may be associated with a certain behavior or pattern during a transaction, and its high importance value suggests that fraudulent behavior is closely related to the anomalous pattern represented by C1. The id\_02 feature (importance: 108.96) also has a high importance in the model and may be closely related to transaction authentication or user behavior. card6 feature (importance: 100.38) represents information related to the payment card, which is particularly important in the identification of fraudulent transactions, as anomalies in the payment method are often a strong warning signal.

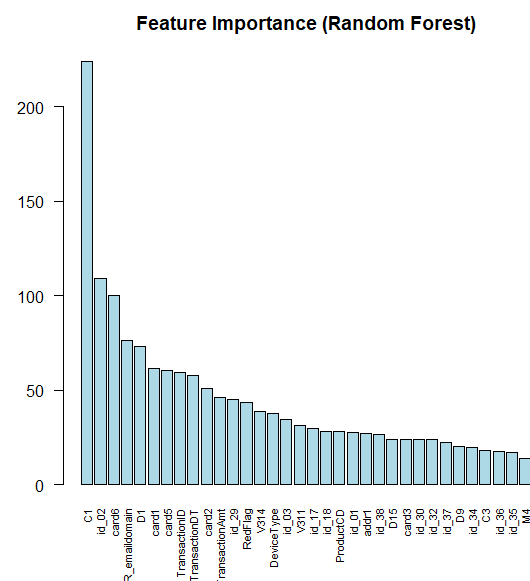


Figure 6 Feature Importance of Random Forest

### 7.2 Red Flag

Recognizing potential risk signals (red flags) is critical in financial fraud prevention. These signals typically exhibit behavioral characteristics that are significantly different from normal transaction patterns and may signal the onset of fraudulent activity. I identified several key risk signals by ranking the importance of features in the dataset. First, feature C1 reflects the association of payment cards with multiple addresses, devices, or IPs, and when frequent, may indicate account misuse. id\_02 features relate to device scoring and login behavior, and anomalies, especially access from high-risk areas, may signal fraud. card6 features with payment card information, such as card type and issuing bank, when present in conjunction with other anomalous features, typically indicate Payment card misuse. The use of temporary or suspicious email domains in the recipient's email domain (R\_emaildomain) is also a risk indicator. multiple transactions within a short period of time in the D1 feature and anomalies in the card1 feature are often indicative of fraudulent activity. Enhancing real-time monitoring and analysis of these features will therefore be a critical step in the effective prevention of financial fraud.

## **8 Conclusion and Recommendations**

In this report, I compare the performance of random forest and neural network models in identifying cases of financial fraudulent transactions, and select the better performing random forest models for feature importance analysis to identify the main risk factors affecting financial fraud. By analyzing several key features (e.g., C1, id\_02, card6, R\_emaildomain, D1, and card1), I found that these features perform particularly well in predicting fraudulent transactions. Therefore, future models can focus on further optimizing the processing of these features by incorporating more high-dimensional information, thus improving the accuracy of fraud detection.

To prevent financial fraud, it's crucial to not just employ data models but also to bolster both corporate governance and internal control measures. It is recommended that while upgrading anti-fraud technology, companies should improve internal controls and strengthen employee training and transaction review mechanisms to more effectively identify and prevent fraud. Furthermore, advocating for inter-departmental cooperation and rapid sharing of information for managing questionable transactions will bolster the comprehensive capability to prevent risks. By enhancing both technological aspects and organizational controls, the Company gains a superior capability to handle financial fraud and refine its system for detection.

## **9 Reference**

- [1] [IEEE-CIS Fraud Detection | Kaggle](#)
- [2] Sandhu N. Red flag behaviors in financial services frauds: a mixed-methods study[J]. Journal of Financial Regulation and Compliance, 2022, 30(2): 167-195.
- [3] Omar N, Johari Z A, Smith M. Predicting fraudulent financial reporting using artificial neural network[J]. Journal of Financial Crime, 2017, 24(2): 362-387.
- [4] Liu C, Chan Y, Alam Kazmi S H, et al. Financial fraud detection model: Based on random forest[J]. International journal of economics and finance, 2015, 7(7).



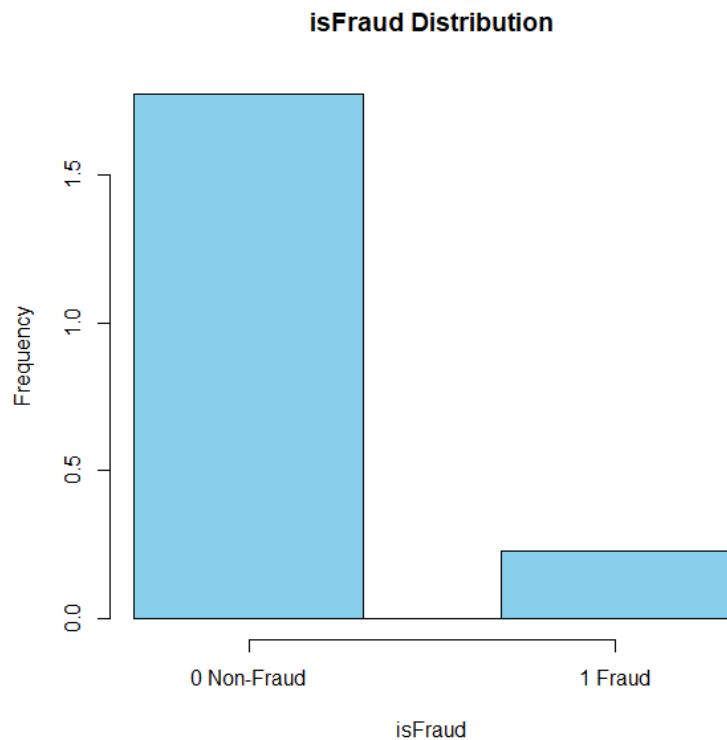
## Appendix

```
> summary(dataset)
TransactionID      isFraud      TransactionDT
Min.   :2987004   Min.   :0.0000   Min.    : 86506
1st Qu.:3078716   1st Qu.:0.0000   1st Qu.: 1900286
Median :3203512   Median :0.0000   Median : 5031202
Mean   :3238037   Mean   :0.1132   Mean    : 6209720
3rd Qu.:3393812   3rd Qu.:0.0000   3rd Qu.:10272711
Max.   :3577534   Max.   :1.0000   Max.    :15811030

TransactionDT..Hour. TransactionAmt      ProductCD
Min.    : 24      Min.    : 0.272   Length:100000
1st Qu.: 527      1st Qu.: 25.419   Class :character
Median :1397      Median : 50.000   Mode  :character
Mean    :1724      Mean    : 83.657
3rd Qu.:2853      3rd Qu.: 100.000
Max.    :4391      Max.    :1800.000

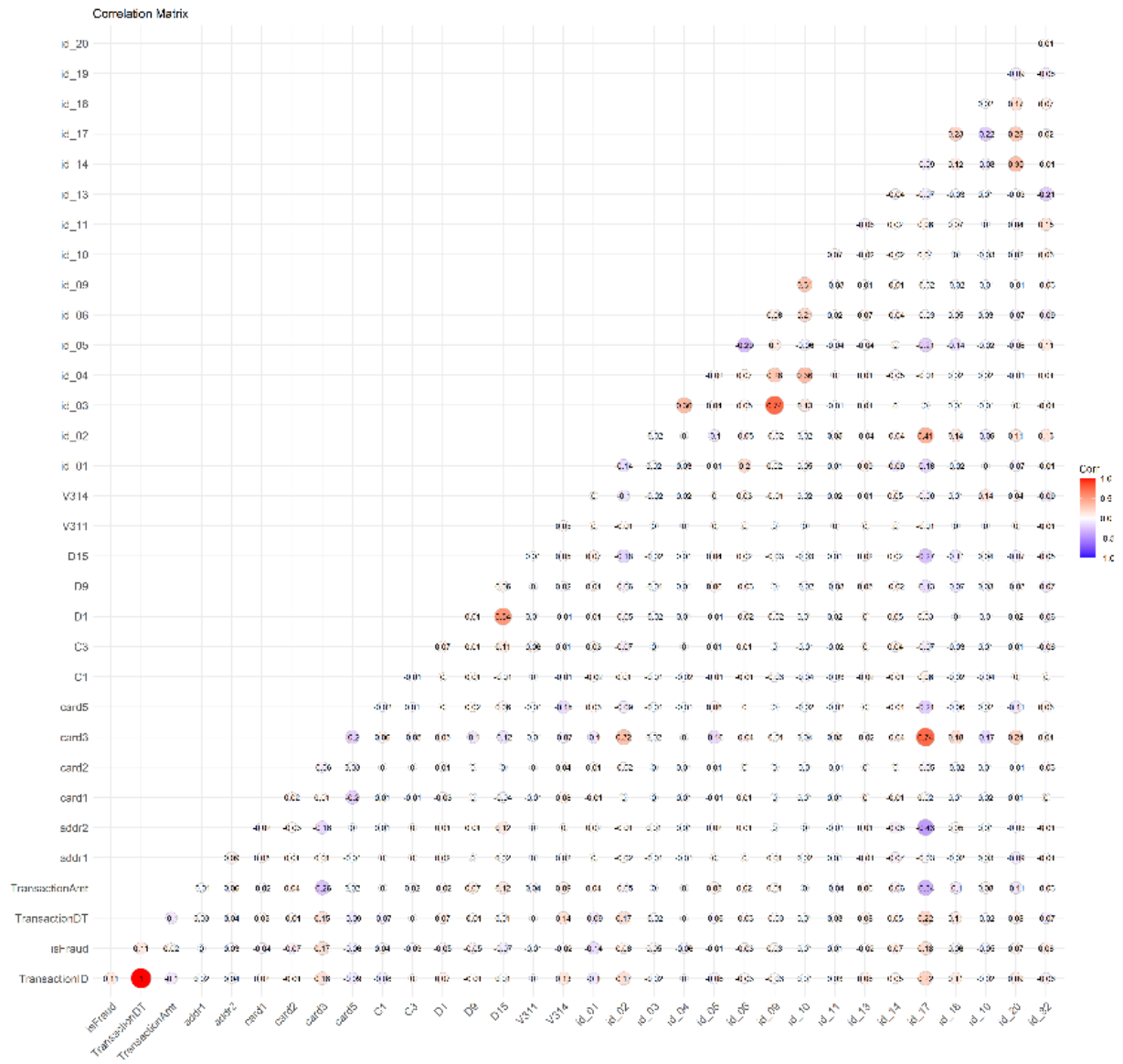
      addr1      addr2      dist1      dist2
Min.    : 82.0   Min.    : 18.00   Mode:logical   Min.    : 0.0
1st Qu.:186.0   1st Qu.: 73.00   NA's:100000    1st Qu.: 7.0
Median :281.0   Median : 73.00                      Median : 38.0
Mean    :278.6   Mean    : 73.01                      Mean    : 230.5
3rd Qu.:312.0   3rd Qu.: 73.00                      3rd Qu.: 207.0
Max.    :522.0   Max.    :110.00                      Max.    :11623.0
NA's    :42881   NA's    :42881                      NA's    :73694
```

Appendix Figure 1 Sample of data summary



Appendix Figure 2 Histogram of isFraud





Appendix Figure 4 Correlation after Feature Selection



Appendix Figure 5 isFraud data after SMOTE

| <p>Confusion matrix:</p> <table border="1"> <thead> <tr> <th></th> <th>0</th> <th>1</th> <th>class.error</th> </tr> </thead> <tbody> <tr> <th>0</th> <td>60754</td> <td>1170</td> <td>0.01889413</td> </tr> <tr> <th>1</th> <td>2370</td> <td>53241</td> <td>0.04261747</td> </tr> </tbody> </table> |       | 0     | 1           | class.error | 0 | 60754 | 1170 | 0.01889413 | 1 | 2370 | 53241 | 0.04261747 | <pre>&gt; print(paste("Accuracy:", accuracy)) [1] "Accuracy: 0.96776050662061" &gt; print(paste("Recall:", recall)) [1] "Recall: 0.979707003512968" &gt; print(paste("F1 Score:", f1_score)) [1] "F1 Score: 0.96995596995597"</pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------------|-------------|---|-------|------|------------|---|------|-------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                      | 0     | 1     | class.error |             |   |       |      |            |   |      |       |            |                                                                                                                                                                                                                                     |
| 0                                                                                                                                                                                                                                                                                                    | 60754 | 1170  | 0.01889413  |             |   |       |      |            |   |      |       |            |                                                                                                                                                                                                                                     |
| 1                                                                                                                                                                                                                                                                                                    | 2370  | 53241 | 0.04261747  |             |   |       |      |            |   |      |       |            |                                                                                                                                                                                                                                     |

Appendix Figure 6 Evaluation of Random Forest

| <p>Confusion Matrix and Statistics</p> <table border="1"> <thead> <tr> <th></th> <th>0</th> <th>1</th> </tr> </thead> <tbody> <tr> <th>0</th> <td>23866</td> <td>2892</td> </tr> <tr> <th>1</th> <td>3261</td> <td>20354</td> </tr> </tbody> </table> |       | 0     | 1 | 0 | 23866 | 2892 | 1 | 3261 | 20354 | <pre>&gt; print(paste("Accuracy:", accuracy)) [1] "Accuracy: 0.877851229825502" &gt; print(paste("Recall:", recall)) [1] "Recall: 0.879787665425591" &gt; print(paste("F1 Score:", f1_score)) [1] "F1 Score: 0.885812378212861" &gt; table(dataset_rf\$isFraud)</pre> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|---|---|-------|------|---|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                       | 0     | 1     |   |   |       |      |   |      |       |                                                                                                                                                                                                                                                                       |
| 0                                                                                                                                                                                                                                                     | 23866 | 2892  |   |   |       |      |   |      |       |                                                                                                                                                                                                                                                                       |
| 1                                                                                                                                                                                                                                                     | 3261  | 20354 |   |   |       |      |   |      |       |                                                                                                                                                                                                                                                                       |

Appendix Figure 7 Evaluation of Neural Network